# 崑山科技大學四年制日間部先進應用材料工程系課程科目表

# Kun Shan University Department of Advanced Applied Materials Engineering
# 4-Year College Curriculum (Day Division)

課程名稱(講授時數－實習時數－學分數)
Course Name (Lecture Hours – Practice Hours – Credits)

111.09.29系課程委員會通過Approved by the Department Curriculum Committee
111.09.29系務會議修正通過Amended and approved by the Department Affairs Meeting
111.11.14院課程委員會通過Approved by the College Curriculum Committee
112.02.13校課程委員會修正通過Amended and approved by the University Curriculum Committe
112.05.15系課程委員會修正通過Amended and approved by the Department Curriculum Committee
112.05.17院課程委員會修正通過Amended and approved by the College Curriculum Committee
112.07.20校課程委員會修正通過Amended and approved by the University Curriculum Committe
112.10.03系課程委員會修正通過Amended and approved by the Department Curriculum Committee
112.11.15院課程委員會修正通過Amended and approved by the College Curriculum Committee
113.01.24校課程委員會修正通過Amended and approved by the University Curriculum Committe
113.04.11系課程委員會修正通過Amended and approved by the Department Curriculum Committee
113.04.26院課程委員會修正通過Amended and approved by the College Curriculum Committee
113.07.30校課程委員會修正通過Amended and approved by the University Curriculum Committe
113.09.25系課程委員會修正通過Amended and approved by the Department Curriculum Committee
113.11.14院課程委員會修正通過Amended and approved by the College Curriculum Committee
114.01.21校課程委員會修正通過Amended and approved by the University Curriculum Committe
114.08.28系課程委員會修正通過Amended and approved by the Departmen Curriculum Committee
114.12.02院課程委員會修正通過Amended and approved by the College Curriculum Committee
115.01.28校課程委員會修正通過Amended and approved by the University Curriculum Committee

112學年度入學適用
Applicable to freshmen enrolled in Fall, 2023

| 第一學年 (1st Academic Year) | | 第二學年 (2nd Academic Year) | | 第三學年 (3rd Academic Year) | | 第四學年 (4th Academic Year) | | |
|---|---|---|---|---|---|---|---|---|
| 第一學期 1st Semester | 第二學期 2nd Semester | 第一學期 1st Semester | 第二學期 2nd Semester | 第一學期 1st Semester | 第二學期 2nd Semester | 第一學期 1st Semester | 第二學期 2nd Semester | 暑期 Summer |
| 共同必修科目(合計 30.5 學分) General Courses (Total：30.5 Credits) | | | | | | | | |
| 體育分組(一)<br>Physical Education (I)<br>2 - 0 - 2 | 體育分組(二)<br>Physical Education (II)<br>2 - 0 - 2 | 生活實踐教育(一)<br>Living Practice (I)<br>0 - 1 - 1 | 服務學習與公民教育<br>Public Service and Civil Education<br>0 - 1 - 0 | 生涯發展與職業倫理<br>Career Development and Work Ethics<br>2 - 0 - 2 | 校訂核心通識-外語選讀***<br>School- Foreign Languages<br>2 - 0 - 2 | | | 英文能力檢定*<br>English Proficiency Examination<br>1 - 0 - 1 |
| 中文閱讀與表達(一)<br>Chinese Reading and Expression(I)<br>2 - 0 - 2 | 勞作教育(一)<br>Labor Education (I)<br>0 - 1 - 0.5 | 校訂分類通識<br>School-designated Liberal Education<br>2 - 0 - 2 | 校訂分類通識<br>School-designated Liberal Education<br>2 - 0 - 2 | 院訂分類通識*****<br>College-designated Liberal Education<br>2 - 0 - 2 | | | | |
| 英文聽力與閱讀(一)<br>English Listening and Reading (I)<br>2 - 0 - 2 | 中文閱讀與表達(二)<br>Chinese Reading and Expression (II)<br>2 - 0 - 2 | 院訂核心通識****<br>College-designated General Core<br>2 - 0 - 2 | - - | - - | - - | - - | - - | - - |
| | 英文聽力與閱讀(二)<br>English Listening and Reading (II)<br>2 - 0 - 2 | 院訂分類通識*****<br>College-designated Liberal Education<br>2 - 0 - 2 | | | | | | |
| | 校訂分類通識<br>School-designated Liberal Education<br>2 - 0 - 2 | | | | | | | |
| 6 - 0 - 6 | 8 - 1 - 8.5 | 6 - 1 - 7 | 2 - 1 - 2 | 4 - 0 - 4 | 2 - 0 - 2 | | | 1 - 0 - 1 |
| 共同選修科目(合計(合計 1 學分) Common Elective (Total：1 Credits) | | | | | | | | |
| | | 外語--第二外語***<br>Foreign Languages (other than English)<br>2 - 0 - 2 | 外語--第二外語***<br>Foreign Languages (other than English)<br>2 - 0 - 2 | | 外語--職場英文***(工商)<br>Foreign Language-Workplace English<br>2 - 0 - 2 | | | 跨領域學習護照**<br>Interdisciplinary Learning Passport<br>1 - 0 - 1 |
| | | | | | | | | 1 - 0 - 1 |

註：Note:
1.* 「英文能力檢定」依「崑山科技大學學生英文基本能力檢定實施辦法」辦理。“English Proficiency Examination” is enacted according to “Kun Shan University Guidelines for English Proficiency Requirement.”
2.** 跨領域學習護照，依「崑山科技大學通識學習護照實施辦法」辦理。“Interdisciplinary Learning Passport” is enacted according to “Kung Shan University Guidelines for General Education Learning Passport.”
3.*** 「外語選讀」為選項式必修科目，課程包括「第二外語」及「職場英文」，擇一門修習即可。“Foreign Languages” compromises “Foreign Languages (other than English)” and “Workplace English”, among which students choose 1 course.

| 院訂必修科目（合計 6 學分）Compulsory Courses of College of Engineering(Total：6 Credits) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 微積分<br>Calculus<br>2 - 0 - 2 | 工程數學<br>Engineering Mathmatics<br>2 - 0 - 2 | | | | | | | |
| 工程技術概論<br>Introduction to Engineering Technology<br>2 - 0 - 2 | | | | | | | | |
| 4 - 0 - 4 | 2 - 0 - 2 | | | | | | | |

| 第一學年<br>(1st Academic Year) | | 第二學年<br>(2nd Academic Year) | | 第三學年<br>(3rd Academic Year) | | 第四學年<br>(4th Academic Year) | | |
|---|---|---|---|---|---|---|---|---|
| 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 暑期<br>Summer |
| 專業必修科目(合計 63 學分) Compulsory Courses (Total: 62 Credits) | | | | | | | | |
| 普通化學(一)<br>General Chemistry (I)<br>3 - 0 - 3 | 普通化學(二)<br>General Chemistry (II)<br>3 - 0 - 3 | 有機化學<br>Organic Chemistry<br>3 - 0 - 3 | 高分子科學<br>Polymer Science<br>3 - 0 - 3 | 材料檢測與分析<br>Materials Testing and Analysis<br>2 - 0 - 2 | 材料檢測與分析實驗<br>Materials Testing and Analysis Experiments<br>0 - 3 - 3 | 學術倫理*<br>Academic Ethics<br>1 - 0 - 1 | 材料研討課程<br>Seminar For Materials<br>0 - 2 - 2 | 暑期校外實習(一)**<br>Summer out of school internship (I)<br>0 - * - 2 |
| 普通物理(一)<br>University Physics(I)<br>2 - 0 - 2 | 普通物理(二)<br>University Physics(II)<br>2 - 0 - 2 | 先進儲能材料<br>Advanced Energy Storage Materials<br>3 - 0 - 3 | 電化學實務<br>Practice of Electrochemical<br>2 - 1 - 3 | 先進複合材料<br>Advanced Composite Materials<br>3 - 0 - 3 | 實務專題(二)<br>Practice and Study on Selected Topic (II)<br>0 - 1 - 1 | 清潔生產<br>Clean Production<br>3 - 0 - 3 | 專業證照***<br>Professional Certificate<br>0 - 1 - 0 | |
| 普通化學實驗<br>General Chemistry Experiments<br>0 - 2 - 2 | 普通物理實驗<br>Physics Experiments<br>0 - 2 - 2 | 材料製備實驗<br>Materials Preparation Experiments<br>0 - 3 - 3 | 半導體製程與設備<br>Semiconductor Process and Equipment<br>3 - 0 - 3 | 材料加工實驗<br>Materials Processing and Experiments<br>1 - 2 - 3 | | | | |
| 節能材料概論<br>Introduction to Energy Conservation<br>3 - 0 - 3 | 半導體材料與元件<br>Elements and Materials of Semiconductor<br>3 - 0 - 3 | | 基礎儀器分析<br>Instrumental Analysis<br>1 - 2 - 3 | 實務專題(一)<br>Practice and Study on Selected Topic (I)<br>0 - 1 - 1 | | | | |
| | | | 計算機程式設計<br>Computer Programming<br>1 - 0 - 1 | | | | | |
| 8 - 2 - 10 | 8 - 2 - 10 | 6 - 3 - 9 | 10 - 3 - 13 | 6 - 3 - 9 | 0 - 4 - 4 | 4 - 0 - 4 | 0 - 3 - 2 | 0 - * - 2 |

註：Note:
1.*「學術倫理」之授課與學分取得依課務組網頁之「學術倫理教育實施辦法」辦理。
The implementation of the “Academic Ethics” course and the acquisition of its credits are carried out in accordance with the Regulations for Academic Ethics Education Implementation on the Academic Affairs Division website.
2.**暑期校外實習課程依材料系之「校外實習必修課程作業要點」辦理。
The summer off-campus internship course is carried out in accordance with the Implementation Guidelines for Off-campus Internship Courses of the Department of Advanced Applied Materials Engineering.
3.***「專業證照」課程依材料系之「專業領域基本能力檢定實施辦法」辦理。
Professional certification courses are carried out in accordance with the Implementation Guidelines for Basic Professional Competency Assessments of the Department of Advanced Applied Materials Engineering.

| 專業選修科目(每學期依實際情況開課) Elective Courses (Based on every semester opening courses) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 材料科學概論(一)<br>Introduction of Materials Science (I)<br>3 - 0 - 3 | 材料科學概論(二)<br>Introduction of Materials Science (II)<br>3 - 0 - 3 | 物理冶金<br>Physical Metallurgy<br>3 - 0 - 3 | 奈米科技<br>Nanoscience<br>3 - 0 - 3 | 綠能材料設計<br>Green Energy Materials Design<br>3 - 0 - 3 | 職業安全與衛生<br>Occupational Safety and Health<br>3 - 0 - 3 | 學年校外實習(一)**<br>Academic year Internship (I)<br>0 - * - 9 | 創意與創業<br>Creativity and Entrepreneurship<br>3 - 0 - 3 | 暑期校外實習(二)<br>Summer out of school internship(II)<br>0 - * - 2 |
| 工業工程概論<br>Introduction of Industrial Engineering<br>3 - 0 - 3 | 材料力學<br>Mechanics of Materials<br>3 - 0 - 3 | 品質管制<br>Quality Control<br>2 - 0 - 2 | 光電材料<br>Optoelectronic Materials<br>3 - 0 - 3 | 織品材料與加工<br>Fabric materials and processing<br>1 - 2 - 3 | 數位印花實務<br>Practice of Digital Printing<br>1 - 2 - 3 | 工廠管理<br>Industry Management<br>2 - 0 - 2 | 學年校外實習(二)**<br>Academic year Internship (II)<br>0 - * - 9 | |
| 工程塑膠<br>Engineering Plastics<br>2 - 0 - 2 | 金屬材料<br>Metallic Materials<br>3 - 0 - 3 | 材料熱力學<br>Thermodynamics of Materials<br>3 - 0 - 3 | 陶瓷材料<br>Ceramic Materials<br>3 - 0 - 3 | 染整實務<br>Practice of Dyeing and Finishing<br>1 - 2 - 3 | 專業證照輔導<br>Professional Counseling Certificate<br>3 - 0 - 3 | 電腦輔助創新設計<br>Innovative Computer Aided Design<br>1 - 2 - 3 | 學期校外實習**<br>Semester Internship<br>0 - * - 9 | |
| 綠色材料<br>Green Materials<br>3 - 0 - 3 | | | 電腦輔助製圖<br>Computer Aided Drawing<br>1 - 2 - 3 | 電腦輔助立體製圖與製造<br>Computer Aided 3D Drawing and Manufacturing<br>1 - 2 - 3 | 半導體產業概論<br>Introduction of Semiconductor Industry<br>3 - 0 - 3 | 太陽光電板材料與檢驗<br>Photovoltaic Panel Materials and Inspection<br>1 - 2 - 3 | 校外職場實習<br>Internship in Workplace<br>0 - * - 7 | |
| | | | | 半導體特用化學品<br>Semiconductor Specialty Chemicals<br>3 - 0 - 3 | 高分子材料特性與加工<br>Polymer Materials Properties and Processing<br>3 - 0 - 3 | 多媒材與色彩計畫運用<br>Applications of Multi-media and Color planning<br>1 - 2 - 3 | | |
| | | | | 高分子化性與物性概論<br>Introduction of Polymer Physics and Chemistry<br>3 - 0 - 3 | | | | |

註：Note:
**暑期校外實習(一)(二)、學期校外實習、學年校外實習(一)(二)等課程依「崑山科技大學學生校外實習辦法」辦理。
The Summer Off-campus Internship (I)(II), Semester Internship, and Academic Year Internship (I)(II) courses are conducted in accordance with the Regulations for Student Off-campus Internships of Kun Shan University.

最低畢業學分： **128** 學分 Minimum for Total Credits: 128 Credits

| 第一學年 (1st Academic Year) | | 第二學年 (2nd Academic Year) | | 第三學年 (3rd Academic Year) | | 第四學年 (4th Academic Year) | | |
|---|---|---|---|---|---|---|---|---|
| 第一學期 1st Semester | 第二學期 2nd Semester | 第一學期 1st Semester | 第二學期 2nd Semester | 第一學期 1st Semester | 第二學期 2nd Semester | 第一學期 1st Semester | 第二學期 2nd Semester | 暑期 Summer |
| 材料應用數位科技微學程(8學分) 註4 Micro Program in Digital Technology for Materials Applications (8 Credits) Note 4 | | | | | | | | |
| | | 通識E General Education E 2 - 0 - 2 | 電腦輔助製圖 Computer Aided Drawing 1 - 2 - 3 | 電腦輔助立體製圖與製造 Computer Aided 3D Drawing and Manufacturing 1 - 2 - 3 | | | | |
| 綠色材料跨院系學分學程課程(16學分)註5 Interdisciplinary Program in Green Materials (16 Credits) Note 5 | | | | | | | | |
| 綠色能源與產業商機 Green Energy and Opportunity 3 - 0 - 3 | 能源應用 Applications of Energy 3 - 0 - 3 | 環境規劃與管理 Environmental Planning and Management 3 - 0 - 3 | 環境影響評估 Environmental Impact Assessment 3 - 0 - 3 | 陶瓷材料 Ceramic Materials 3 - 0 - 3 | 綠色材料 Green Materials 3 - 0 - 3 | 材料檢測與分析 Materials Testing and Analysis 2 - 0 - 2 | 暑期校外實習(一) Summer out of school internship(I) 0 - * - 2 | 學期校外實習 Semester Intership 0 - * - 9 |
| 學年校外實習(一)(二) Academic year Internship (I)(II) 0 - * - 18 | | | | | | | | |

註：Note:

1.本系最低畢業學分為128學分。

The minimum number of credits required for graduation in this department is 128.

2.暑期校外實習(必修)實習時數至少160小時；學期校外實習 (選修)開設共9學分為連續4.5個月之實習，學年校外實(一)(二) (選修)開設共18學分為連續9個月之實習，相關辦法另訂之。

The Summer Off-campus Internship (compulsory) requires a minimum of 160 hours. The Semester Internship (elective) offers 9 credits and consists of a continuous internship period of 4.5 months. The Academic Year Internship (I)(II) (elective) offers 18 credits and consists of a continuous internship period of 9 months. Relevant regulations are stipulated separately.

3.選修學期校外實習或學年校外實習(一)者得抵免暑期校外實習(一) 2學分且取得該專業選修校外職場實習7學分。

Students who take the Elective Semester Internship or Academic Year Internship (I) may be granted an exemption of 2 credits from Summer Off-campus Internship (I) and earn 7 credits for the Elective Professional Workplace Internship.

4.學生須修畢本系材料應用數位科技微學程課程始得畢業。

Students in this department must complete the Micro Program in Digital Technology for Materials Applications and obtain the corresponding certificate before graduation.

5.學生須修畢本院跨院系學程或本校他院之跨院系學程，始得畢業。

Students in this department must complete an interdisciplinary program offered by the College of Engineering or a cross-departmental program offered by another college of the university in order to graduate.